

Hrvoje Jerković, PhD

Data Architect | Principal Data Engineer

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Online CV & detailed project portfolio

hrvoje.github.io/cv/

Profile

Data and solution architect combining hands-on engineering, technical leadership and education. Data and ML project experience since 2017, with a broader background in information systems. Specialises in lakehouse architecture, data lineage, reliable data pipelines and practical AI applications across industries, including automation, predictive analytics and data modernisation.

Core expertise

- **Architecture:** lakehouse/DWH, Medallion, relational and dimensional modelling; platform transformation, upgrades and modernisation, cost-efficient migrations, performance and cost optimisation.
- **Business and data consulting:** understanding business processes and data at source; tracing meaning, transformations and root causes with clients, then shaping architecture, process redesign, migrations and new services.
- **Engineering:** Python, SQL, dbt, Spark, Delta Lake, Airflow, Kafka, Snowflake, PostgreSQL, Oracle, Elasticsearch and OpenSearch; batch, streaming, APIs and CDC patterns. NoSQL: Amazon DynamoDB and MongoDB.
- **ETL/ELT and ingestion:** Matillion, Oracle Data Integrator (ODI), Fivetran; serverless pipelines with AWS Lambda, AWS Step Functions and AWS Glue.
- **Cloud and delivery:** AWS, Google Cloud (GCP), Microsoft Azure, Databricks, DigitalOcean, Oracle Cloud, Docker, Git and CI/CD.
- **Analytics and digital marketing:** Google Analytics, Google Ads (AdWords), Google AdSense, Google Search Console (Webmaster Tools), Facebook/Meta marketing, cross-channel data integration, Tableau, Power BI and DAX.
- **Data governance:** AWS Glue Crawlers and AWS Glue Data Catalog for schema discovery, schema-change tracking and metadata tables; Collibra, OpenMetadata, Unity Catalog, table/column lineage, data quality, access controls and auditability.
- **Delivery and applied AI:** prototypes and demos, implementation planning, technical coordination, testing and handover; RAG for document retrieval, local LLM evaluation, automation, predictive analytics and AI-assisted migration; mentoring and workshops.

Selected experience

Data Architect / Principal Data Engineer

Raiffeisenbank Austria d.d., Croatia (RBA) | Sep 2025 - Feb 2026 | Project case study

- Designed and implemented a banking lakehouse migration framework for an Oracle/PostgreSQL estate of approximately 4,300 ETL mappings and 1,800 landing tables. Defined Bronze/Silver/Gold layers and reusable dbt models targeting Spark and Delta Lake.
- Used Oracle Data Integrator (ODI) extensively in RBA reengineering. Automated analysis of ODI, DataStage and PL/SQL flows to reconstruct table/column lineage across 20+ source systems, with dependency graphs and structured metadata.
- Built orchestration with nine specialised AI agents to translate legacy logic into dbt SQL. Ordered models by their dependencies, compiled and ran the SQL, compared results with production data and escalated failed checks.
- Removed the need for approximately 340 separately stored data-quality tables by running quality checks within the new processing flow. Handed over reusable models and pipelines to the delivery team.

Data Architect / Principal Data Engineer / Data Consultant

Intuita Consulting (UK group) | Sep 2022 - Feb 2024

- VIOOH / JCDecaux: led SQL/dbt upgrades in Kafka/S3/Airflow/Snowflake advertising analytics processing 1–5 million nested JSON records/hour; investigated issues with DataDog and Grafana.
- Designed and implemented a pipeline anomaly-detection and notification application using Python, SQL, AWS Athena, Lambda, Glue and Snowflake integrations.
- Altus Group: used Matillion to ingest data from Google Analytics, email, Salesforce, advertising and social sources; unified analytics in Snowflake with dbt and SQL.
- Built DataDQ advertising-data workflows using Fivetran for ingestion and initial transformations, plus Airflow, S3 and Redshift for file loads, staging and viewability updates.
- Comcast / Sky — metadata engineering: used Python and PostgreSQL to structure table, column, business and governance metadata from Confluence for analysis and Collibra.
- Comcast / Sky — GCP/AlloyDB prototype: built and tested SQL/PL/pgSQL schema-metadata capture, refresh after DDL changes and table/column audit records.
- Hyperoptic: mapped telecom processes, entities and attributes to TM Forum SID/ODA using ER/UML models.

Further architecture, AI and education experience — most recent first

Solution Architect / Data Architect / Technical Lead - eADR

Agency for Commercial Activities (AKD) / Croatian Ministry of the Sea, Transport and Infrastructure | Nov 2025 - Aug 2026

- Designed architecture and development specifications for six modules of an EU-funded national road-transport platform, connecting business processes, domain models, APIs, identity/RBAC, audit trails and reporting.
- Led technical coordination, stakeholder decisions and UAT with requirements-to-test traceability. Built a local RAG knowledge base for project specifications using semantic search, BM25 and MCP.
- AFFINIS unified analytics: designed the lakehouse expansion for eADR and eFTI/NSCP data; delivered data-model, pipeline and reporting specifications, analytical-validation requirements and an infrastructure-upgrade assessment.

Data Engineering Instructor / Workshop Author

NEOS (KONČAR Group) | Apr 2026

- Designed and delivered a two-day Azure Databricks engineering workshop covering Delta Lake, Auto Loader, Structured Streaming, SQL, Unity Catalog, data quality and orchestration.
- Built banking demos, guided labs and assignments with dbt tests, Bundles and delivery workflows. Linked exercises to Associate/Professional certification topics and adapted them to the team's environment.

Lead Data Architect / Project Manager - Technical Bid and Demo

NEOS / Central Bank of Oman — pre-contract evaluation | Sep 2025 - Feb 2026

- Led on-premises lakehouse design, technical coordination and end-to-end requirements analysis, from real-time mobile-banking scenarios to big-data analytics. Prototyped data-lake exploration, regulatory submissions, validation-rule configuration, version comparisons and payment dashboards using simulated data; designed batch/streaming ingestion, governance and analytical access.
- Reached final-stage evaluation with the NEOS team and presented my work and project references to the bank's delegation in Zagreb.

Data Architect / Technical Lead - Road Transport Analytics

Agency for Commercial Activities (AKD) / Croatian Ministry of the Sea, Transport and Infrastructure | Jul - Dec 2024 | AKD Affinis

- Designed an open-source big-data lakehouse architecture and analytics platform sized for 250 TB+, 75,000+ drivers and 65,000+ vehicles. Defined 40+ safety, compliance and efficiency indicators integrating tachograph, inspection and ministry data.
- Prepared infrastructure, processing and other specifications in the full implementation documentation: Spark, MinIO, Trino, StarRocks, dbt, Dagster, OpenMetadata, PostgreSQL, Superset and Kubernetes.
- Led detailed task and specification development for data engineers, DevOps and analytics specialists; evaluated candidates and defined delivery responsibilities.

Data Architect / Head of AI

8tracks.com / mixer.fm | Apr - Aug 2024

- Big-data processing and analysis of audio files: developed AWS components using S3, Lambda, EFS, Amazon DynamoDB (NoSQL), PostgreSQL/RDS, SAM, CloudFormation and CodeBuild. Integrated pretrained models for genre, style, mood, instrumentation and danceability; built feature extraction, metadata processing and a waveform demo. Designed recommendation services with DynamoDB for user interactions and Redshift for analytical history. Evaluated Elasticsearch and OpenSearch for audio-feature and metadata search.

Data Engineer / Data Scientist / Lecturer

Algebra / Algebra LAB | Mar 2018 - Sep 2022

- IRI2 / Neural Grid: real-time grid regulation; developed software to compare phasor angles from PMU measurements, perform power-system calculations and trigger PLC actions. Used Python, Java and Kafka Streams; achieved a response below 50 ms, significantly faster than SCADA in this project.
- National labour-market analytics: served as Data Architect / Lead Data Engineer, migrated analytical processing to SQL Server and integrated employment and education data.
- Agricultural analytics: developed complex SQL views and Tableau dashboards for ministry-wide analysis of agricultural data.
- Erasmus ADSEE: developed computer-vision demonstrations and educational material using Python, OpenCV and Jupyter, connecting video preparation, detections and analytical tables.
- Course leader for Machine Learning on the Data Science master's, delivering part of its lectures and practical exercises; also taught Disruptive Technologies and E-business.

Earlier experience: Zagreb School of Economics and Management, 2002-2017 - information systems development and integration, technical leadership, digital learning and professional education.

Additional project work

- Vehicle-market analytics and price prediction: developed an XGBoost car-price model for Oglasnik.hr, with Tableau and Power BI analytics. Built an independent Airflow/PostgreSQL collection platform with new-ad detection and price-change tracking.
- Desktop property analytics: developed a Python/PyQt5 application for collecting, filtering and analysing property listings, with configurable notifications.
- News portal: built Python/PostgreSQL collection, Dagster jobs, AI summaries and article/category publishing; developed the project using Oracle Cloud.
- National register of regulated professions: designed the SQL Server database and developed a C#/ASP.NET Core application for profession records, qualification-recognition information, search, editing and audit.
- Occupational classifications: developed SQL Server and C#/ASP.NET Core software for maintaining Croatian occupational codes and their relationships with the Croatian Qualifications Framework, including search and change history.
- eFTI portal: developed Java/Spring Boot and React/TypeScript software for freight datasets, XML validation, subset selection, access links and audit.
- Independent websites and digital marketing: built and maintained websites, configured analytics and marketing tools, and taught these skills through practical student work.

Education and professional development

PhD in Applied Computer Science — University of Zagreb, Faculty of Electrical Engineering and Computing

Awarded 2012. Research: information-system interoperability and integration of learning platforms with digital repositories.

Udacity: Machine Learning Engineer Nanodegree; Deep Learning Nanodegree.

Credentials: SnowPro Core (2024); AWS Cloud Practitioner (2022).

Languages: English - full professional proficiency, including consulting and teaching; Croatian - native.